

ASSIGNMENT 5

Case Study Practical

Task (1):

Laptop Won't Power On – 5 Troubleshooting Steps:

- 1. Check the charger:** Make sure the charger is properly connected and the charging light turns on.
- 2. Check the power socket:** Try another working socket to make sure the power source is okay.
- 3. Check the battery:** Look for a dead or faulty battery. If possible, test with the charger connected without the battery.
- 4. Perform a power reset:** Disconnect the charger, remove the battery if possible, hold the power button for 15–20 seconds, then reconnect power and try again.
- 5. Check hardware:** If there are still no lights, fan sounds, or display, check for a faulty power port, motherboard, or other internal hardware.

Task (2):

Black Screen, CPU Running:

- 1. Check monitor power:** Make sure the monitor is switched on and the power cable is connected.
- 2. Check display cable:** Check the HDMI/DisplayPort/VGA cable and reconnect it firmly.
- 3. Check input source:** Select the correct HDMI/DisplayPort input on the monitor.
- 4. Check PC connections:** Make sure the GPU is properly seated and its power cables are connected.
- 5. Try another cable/port:** Use a different display cable or port to find a faulty connection.
- 6. Check display settings:** Press Windows + P and select Duplicate or Extend.
- 7. Restart the PC:** Check whether the display returns after restarting.

Task (3):

Wireless Mouse Troubleshooting:

1. Check the mouse:

- Make sure the mouse is switched ON.
- Replace or recharge the battery.

2. Check the USB receiver:

- Remove and reconnect the wireless receiver.
- Try the receiver in a different USB port.

3. Test the USB port:

- Connect another USB device, such as a pen drive.

4. Check system settings:

- Open Device Manager and check Mice and other pointing devices.
- Look for a warning or driver problem.

5. Test the mouse on another PC:

- If it does not work there, the mouse/receiver is faulty.
- If it works on another PC, the problem is likely with the original PC's USB port, driver, or settings.

Task (4):

Gaming PC Overheating – 3 Hardware Causes:

1. Faulty or slow fan:

- **How to Check:** Check if the CPU/GPU fans are spinning properly. Replace a fan if it is not working.

2. Dust buildup:

- **How to Check:** Open the PC and check the fans, heatsink, and air vents for dust. Clean them carefully with compressed air.

3. Old/dry thermal paste:

- **How to Check:** Check the thermal paste between the CPU and heatsink. If it is dry or damaged, clean it and apply new thermal paste.
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